

MATH3303 – Lecture 28

Gaussian Integers; gcds, Associates, and the Euclidean Algorithm

Review Notes

Reference: Lee, Abstract Algebra, Section 10.2

Overview

This lecture works out the canonical nontrivial example of a Euclidean domain – the Gaussian integers $\mathbb{Z}[i]$ – and then develops the surrounding theory of divisibility.

- **Gaussian integers** $\mathbb{Z}[i]$: full proof that $\varepsilon(a + bi) = a^2 + b^2$ is a Euclidean function
- **Non-example:** $\mathbb{Z}[x]$ is not a Euclidean domain
- **Divisibility, common divisor, gcd** in commutative rings
- **Associates** and the mutual-divisibility lemma
- **Euclidean algorithm** and a Bézout-style identity $au + bv = d$
- **Uniqueness of gcd up to associates**

Connection to Lecture 27. Lecture 27 introduced the definition of a Euclidean domain and verified $\mathbb{Z}$, fields, and $K[x]$. The Gaussian integers $\mathbb{Z}[i]$ are a more sophisticated example – one whose Euclidean function involves nontrivial geometry. The rest of the lecture pins down the abstract notions of divisibility, gcd, and associates that we will use throughout Chapter 10.

1 The Gaussian Integers

Example 1.1: Gaussian integers $\mathbb{Z}[i]$ (from lecture; cf. Lee, Example 10.8)

Let

$$R = \mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}.$$

Define $\varepsilon : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$ by

$$\varepsilon(a + bi) := a^2 + b^2.$$

Then (R, ε) is a Euclidean domain.

Step 1: ε is multiplicative. For all $a, b, c, d \in \mathbb{Z}$,

$$\begin{aligned} \varepsilon((a + bi)(c + di)) &= \varepsilon((ac - bd) + (ad + bc)i) \\ &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &= (a^2 + b^2)(c^2 + d^2) \\ &= \varepsilon(a + bi) \cdot \varepsilon(c + di). \end{aligned}$$

In particular, $\varepsilon((a + bi)(c + di)) \geq \varepsilon(a + bi)$, verifying condition (1).

Step 2: Division with remainder. Let $u = a + bi$, $v = c + di$ be nonzero in R . We aim to write $u = qv + r$ with $r = 0$ or $\varepsilon(r) < \varepsilon(v)$.

Work in $\mathbb{C}$:

$$uv^{-1} = \frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i =: s + ti,$$

where $s, t \in \mathbb{Q}$.

Choose integers m, n such that

$$|s - m| \leq \frac{1}{2}, \quad |t - n| \leq \frac{1}{2}.$$

Then

$$uv^{-1} = (m + ni) + ((s - m) + (t - n)i) \implies u = (m + ni)v + ((s - m) + (t - n)i)v.$$

Set $q := m + ni \in \mathbb{Z}[i]$ and $r := u - qv \in \mathbb{Z}[i]$. To verify the remainder condition,

$$\begin{aligned} \varepsilon(r) &= \varepsilon(((s - m) + (t - n)i) \cdot v) \\ &= \varepsilon((s - m) + (t - n)i) \cdot \varepsilon(v) \\ &= ((s - m)^2 + (t - n)^2) \cdot \varepsilon(v) \\ &\leq \left(\frac{1}{4} + \frac{1}{4}\right)\varepsilon(v) = \frac{1}{2}\varepsilon(v) < \varepsilon(v). \end{aligned}$$

Hence $u = qv + r$ with $\varepsilon(r) < \varepsilon(v)$. □

Remark 1.1: A non-example: $\mathbb{Z}[x]$ (from lecture)

$\mathbb{Z}[x]$ is *not* a Euclidean domain. In fact it is not even a PID – this is the focus of Lecture 29. The obstruction is that $\mathbb{Z}$ is not a field, so the division algorithm in $\mathbb{Z}[x]$ fails for polynomials whose leading coefficient is not ± 1 .

2 Divisibility, Common Divisors, and gcds

Definition 2.1: Divisibility, common divisor, gcd (from lecture; cf. Lee, Defs. 10.5, 10.6)

Let R be a commutative ring. For $a, b \in R$, we say:

- a divides b , written $a \mid b$, if $\exists c \in R$ such that $b = ac$.
- $c \in R$ is a *common divisor* of a and b if $c \mid a$ and $c \mid b$.

If (R, ε) is a Euclidean domain, $d \in R$ is a *greatest common divisor (gcd)* of a and b if

- d is a common divisor of a and b ; and
- whenever $c \in R$ satisfies $c \mid a$ and $c \mid b$, we have $\varepsilon(c) \leq \varepsilon(d)$.

Remark 2.1: gcds are not unique (from lecture)

- In $\mathbb{Z}$, $\gcd(5, 10) \in \{5, -5\}$. We conventionally take the positive gcd.
- In $K[x]$ (with K a field), if $d(x)$ is a gcd then so is $ad(x)$ for any $a \in K^\times$. We conventionally take the *monic* gcd.

3 Associates

Definition 3.1: Associates (from lecture; cf. Lee, Def. 10.8)

Let R be a commutative ring with identity. Then $a, b \in R$ are *associates* if there is a unit $u \in R$ such that $a = bu$.

Remark 3.1: (from lecture)

“Being associates” is an equivalence relation on R .

Example 3.1: (from lecture)

- $\mathbb{Z}^\times = \{1, -1\}$, so $a, b \in \mathbb{Z}$ are associates iff $a = \pm b$.
- $K[x]^\times = K^\times$ (with K a field), so $f, g \in K[x]$ are associates iff $f = ug$ for some $u \in K^\times$.

Lemma 3.1: Associates via mutual divisibility (from lecture; cf. Lee, Lemma 10.1)

Let R be an integral domain and $a, b \in R \setminus \{0\}$. Then

$$a, b \text{ are associates} \iff a \mid b \text{ and } b \mid a.$$

Proof. $(\Rightarrow)$ If $a = bu$ for $u \in R^\times$, then $b \mid a$. Also $b = au^{-1}$, so $a \mid b$.

$(\Leftarrow)$ Suppose $b = ax$ and $a = by$ for some $x, y \in R$. Then $a = by = axy$, so $a(1 - xy) = 0$. Since $a \neq 0$ and R is an integral domain, $xy = 1$. Hence x, y are units, so a, b are associates. $\square$

4 The Euclidean Algorithm

Theorem 4.1: Euclidean Algorithm (from lecture; cf. Lee, Thm. 10.4)

Let R be a Euclidean domain and $a, b \in R$ with $b \neq 0$. If $b \mid a$, then b is a gcd of a and b . Otherwise, write

$$\begin{aligned} a &= bq_1 + r_1, & \varepsilon(r_1) &< \varepsilon(b), \\ b &= r_1q_2 + r_2, & \varepsilon(r_2) &< \varepsilon(r_1), \\ r_1 &= r_2q_3 + r_3, & \varepsilon(r_3) &< \varepsilon(r_2), \\ &\vdots \\ r_{k-2} &= r_{k-1}q_k + r_k, & \varepsilon(r_k) &< \varepsilon(r_{k-1}), \\ r_{k-1} &= r_kq_{k+1} + 0, \end{aligned}$$

where $q_i, r_i \in R$ and $r_i \neq 0$. Then r_k is a gcd of a and b .

Proof. Exercise. □

Lemma 4.1: Bézout-style identity (from lecture; cf. Lee, Cor. 10.3)

Let d be the gcd of a and b found in Theorem 4.1. Then there exist $u, v \in R$ such that

$$au + bv = d.$$

Proof. If $b \mid a$, then $d = b = a(0) + b(1)$.

Otherwise, from the second-to-last equation,

$$d = r_k = r_{k-2} + r_{k-1}(-q_k).$$

From $r_{k-3} = r_{k-2}q_{k-1} + r_{k-1}$,

$$d = r_{k-2} + (r_{k-3} - r_{k-2}q_{k-1})(-q_k).$$

Keep substituting backwards. Eventually d is a multiple of a plus a multiple of b . □

Remark 4.1: (from lecture; cf. Lee, Cor. 10.4)

If c is a common divisor of a and b , then $c \mid d$ by Lemma 4.2 (since $d = au + bv$).

Theorem 4.2: Uniqueness of gcd up to associates (from lecture; cf. Lee, Thm. 10.5)

Let (R, ε) be a Euclidean domain. Let $a, b \in R$, not both zero, and let d be any gcd of a and b . Then $c \in R$ is a gcd of a and b iff c and d are associates.

Proof. ($\Rightarrow$) Let g be the gcd found by the Euclidean Algorithm. If we show c is an associate of g , and d is an associate of g , then by transitivity c and d are associates.

Since c is a common divisor of a and b , $c \mid g$ (Remark 4.3). Also g is a common divisor of a and b , and c is a gcd, so $\varepsilon(g) \leq \varepsilon(c)$. But from $c \mid g$ and $g \neq 0$, $\varepsilon(c) \leq \varepsilon(g)$. Hence $\varepsilon(c) = \varepsilon(g)$, so $g \mid c$ as well (otherwise dividing g by c would give a smaller-norm common divisor). By

Lemma 3.4, c and g are associates. The same argument applied to d in place of c gives that d and g are associates.

($\Leftarrow$) Let c and d be associates. Since $d \mid a$ and $d \mid b$, also $c \mid a$ and $c \mid b$. Since $c \mid d$ and $d \mid c$, $\varepsilon(c) = \varepsilon(d)$. Hence c is a gcd of a and b . $\square$

Remark 4.2: Independence of ε (from lecture)

By Theorem 4.4, the set of gcds in a Euclidean domain does not depend on the particular Euclidean function used.

Checklist of Skills

- ☐ Prove that $\mathbb{Z}[i]$ with norm $\varepsilon(a + bi) = a^2 + b^2$ is a Euclidean domain.
- ☐ Carry out division-with-remainder in $\mathbb{Z}[i]$ via the “round to nearest lattice point” construction.
- ☐ Cite the non-example: $\mathbb{Z}[x]$ is not a Euclidean domain.
- ☐ State the definitions of divisibility, common divisor, gcd, and associates.
- ☐ Prove that two nonzero elements of an integral domain are associates iff they divide each other.
- ☐ State the Euclidean algorithm and the Bézout-style identity $au + bv = d$.
- ☐ Use the Euclidean algorithm to compute gcds and express them as linear combinations.
- ☐ Prove that any two gcds in a Euclidean domain are associates.